

+0/1/60+

Probabilistic Artificial Intelligence
Questions Pack

February 4, 2022

Time limit: 120 minutes

Instructions. This pack contains all questions for the final exam. It contains the questions only. Please use the accompanying answer sheet to provide your answers by blackening out the corresponding squares. As the exam will be graded by a computer, please **make sure to do blacken out the whole square and do not use ticks or crosses**. *During* the exam you can use a **pencil** to fill out the squares as well as an **eraser** to edit your answers. *After* the exam is over, we will collect the questions pack and provide you with additional time to blacken out the squares on the answer sheet with a **black pen**. *Nothing* written on pages of the question pack will be collected or marked. **Only the separate answer sheet with the filled squares will be marked**.

Please make sure that your answer sheet is clean and all answers are clearly marked by filling the squares out completely. We reserve the right to classify answers as wrong without further consideration if the sheet is filled out ambiguously.

Collaboration on the exam is strictly forbidden. You are allowed a summary of *two* A4 pages and a simple, non-programmable calculator. The use of any other helping material or collaboration will lead to being excluded from the exam and subjected to disciplinary measures by the ETH Zurich disciplinary committee.

Question Types In this exam, you will encounter the following question types.

- **Multiple Choice questions with a single answer.**

  Multiple Choice questions have **exactly one** correct choice. Depending on the difficulty of the question **2, 3, or 4** points are awarded if answered correctly, and **zero points** are awarded if answered wrong or not attempted.

- **True Or False questions.**

  Each True Or False questions has a value of **1 point** if answered correctly, **0 points** if answered wrong or not attempted.

There are no negative grades in this exam. Please also note that **not** all questions need to be answered correctly to achieve the best grade.

All questions in the same section might be related. In addition, some questions require information from previous parts. Hence, if you intend to skip some questions, make sure that you read all notes and details between the questions.

The notation $\mathcal{N}\left(\mu, \sigma^2\right)$ will designate a Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}\left(3, 4\right)$ designates the Gaussian distribution with mean 3 and standard deviation 2.

Good luck!

For your examination, preferably print documents compiled from auto-multiple-choice.

1 Probability, Bayesian Linear Regression

1.1 Linear dynamical system

Consider a linear dynamical system where the state at time $k+1$, $x_{k+1} \in \mathbb{R}$, is given by

$$x_{k+1} = ax_k + \varepsilon_{k+1}.$$

Here, $a \in \mathbb{R}$ is an unknown parameter and ε_{k+1} is an unobserved i.i.d. noise with distribution $\varepsilon_{k+1} \sim \mathcal{N}\left(0, \sigma^2\right)$. Suppose the noise standard deviation σ is known, and at the beginning ($k = 0$), the initial state of the system is given by $x_0 = 0$. We then observe the states $x_1, x_2, \ldots, x_n$.

Question 1 *(4 points)* What is the maximum likelihood estimator $\widehat{a}$ for the unknown parameter a based on the model described above?

A $\widehat{a} = \frac{\sum_{i=1}^{n-1} x_{i+1}x_{i-1}}{\sum_{i=0}^{n-1} x_i^2}$

B $\widehat{a} = \frac{\sum_{i=0}^{n-1} x_i^2}{\sum_{i=0}^{n-1} x_{i+1}x_i}$

C $\widehat{a} = \frac{\sum_{i=0}^{n-1} x_{i+1}x_i}{\sum_{i=0}^{n} x_i^2}$

D $\widehat{a} = \frac{\sum_{i=0}^{n-1} x_{i+1}x_i}{\sum_{i=0}^{n-1} x_i^2}$

For the next question, we take a Bayesian approach and assume a prior distribution for a. Accordingly, suppose $a \sim \mathcal{N}\left(0, \frac{\sigma^2}{\lambda}\right)$ for some $\lambda > 0$. We then wish to compute the posterior distribution of a given the observations, i.e. $p(a|x_0, \ldots, x_n)$.

Question 2 *(3 points)* What is the variance of the posterior distribution $\mathrm{var}(a|x_0, \ldots, x_n)$?

A $\frac{\sigma^2}{\lambda + \sum_{i=0}^{n-1} x_{i+1}x_i}$

B $\frac{\sigma^2 + \sum_{i=0}^{n-1} x_i^2}{\lambda}$

C $\frac{\sigma^2}{\lambda + \sum_{i=0}^{n} x_i^2}$

D $\frac{\sigma^2}{\lambda + \sum_{i=0}^{n-1} x_i^2}$

1.2 Short questions

Question 3 *(1 point)* Let $X_1, \ldots, X_n$ be binary random variables with the joint distribution $P(X_1 = x_1, \ldots, X_n = x_n)$. To completely specify $P(X_1 = x_1, \ldots, X_n = x_n)$, we need $\mathcal{O}(n^2)$ parameters in the worst case.

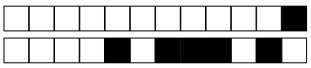
A True B False

Question 4 *(1 point)* Let $(X, Y)^T \sim \mathcal{N}\left((\mu_x, \mu_y)^T, \Sigma\right)$, such that X and Y are uncorrelated, i.e. the off-diagonal elements of Σ are zero. It is possible that X and Y are not independent.

A True B False

Question 5 *(1 point)* Let X, Y be random variables (not necessarily independent) defined over the same probability space, and assume both have the same marginal distribution $\mathcal{N}(0, 1)$. We can then conclude that for any $a, b \in \mathbb{R}$, $aX + bY$ follows a Gaussian distribution.

A True B False



Question 6 *(1 point)* Let X, Y, Z be i.i.d random variables distributed as $\mathcal{N}(0, 1)$. Then the random variable $X \sin(Z) + Y \cos(Z)$ is also Gaussian. (Hint: condition on Z.)

A True B False

Question 7 *(1 point)* Aleatoric uncertainty represents the uncertainty about the model due to the lack of data.

A True B False

2 Gaussian Processes

We are given a dataset $\mathcal{D} = \{(x_1, y_1), \dots (x_{10}, y_{10})\}$ consisting of 10 observation pairs $x, y \in \mathbb{R}$, as depicted in Figure 1. We assume every data point comes from the model

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \overset{\text{i.i.d}}{\sim} \mathcal{N}\left(0, \sigma^2\right),$$

where f is an *unknown* function and σ is a *known* parameter. Due to the small dataset, uncertainty quantification is very important and we decide to model f with Gaussian Processes (GPs).

Question 8 *(2 points)* We consider four different models, each described by $f \sim GP(0, k)$ with a different kernel function k. We consider the following kernel functions:

(A) $k(x_1, x_2) = l^2 x_1 x_2$,

(B) $k(x_1, x_2) = e^{-\frac{(x_1 - x_2)^2}{l^2}}$,

(C) $k(x_1, x_2) = e^{-\frac{|x_1 - x_2|}{l^2}}$,

(D) $k(x_1, x_2) = e^{-\frac{2\sin^2(\pi|x_1 - x_2|/p)}{l^2}}$,

and for each kernel, the hyper-parameters $l, p > 0$ are fitted by maximizing the marginal log-likelihood of the data. Please match each kernel to its corresponding plot depicted in Figure 1.

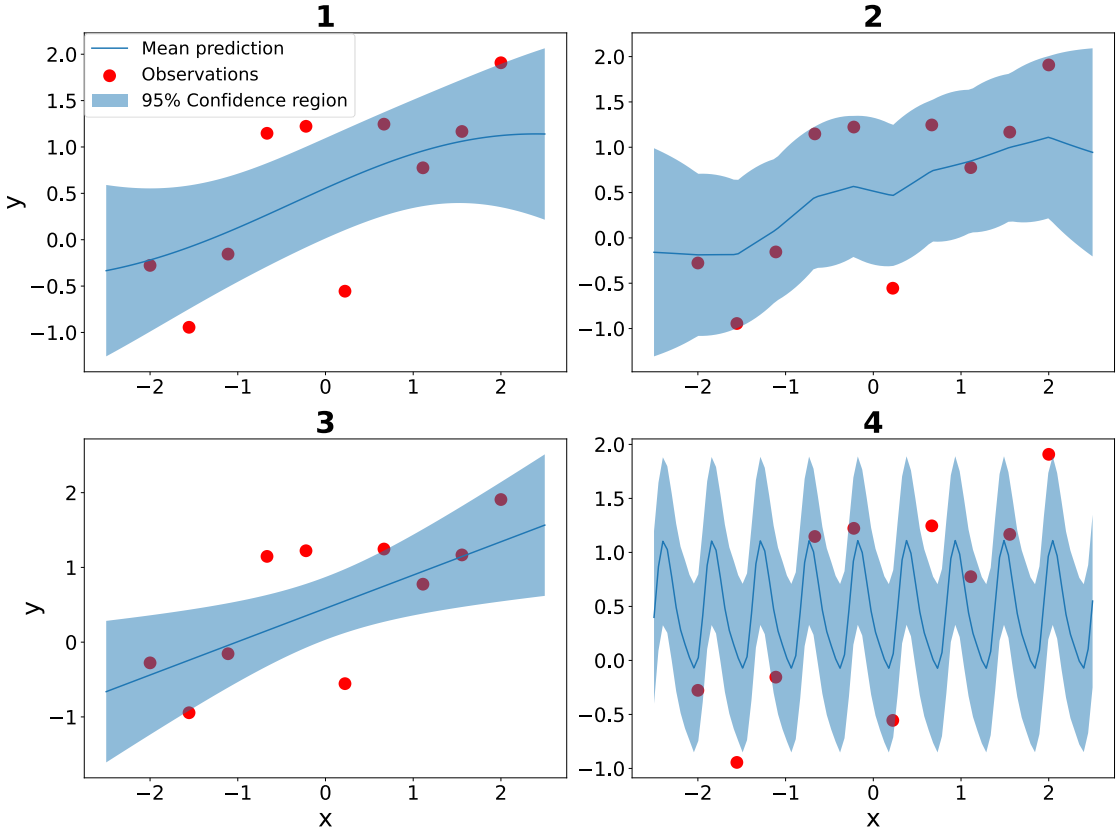


Figure 1: Dataset with 4 GP predictions of mean and 95% confidence region around the mean.

A 1A, 2B, 3C, 4D	C 1C, 2B, 3A, 4D	E 1A, 2C, 3B, 4D
B 1B, 2C, 3D, 4A	D 1C, 2B, 3A, 4D	F 1B, 2C, 3A, 4D

After we look at Figure 1, we decide to use a *linear* kernel, i.e., $k(x_1, x_2) = l^2 x_1 x_2$. For the rest of this exercise, assume $k(x_1, x_2) = l^2 x_1 x_2$. We further define $\boldsymbol{x} = (x_1, \dots, x_{10})^\top$ and $\boldsymbol{y} = (y_1, \dots, y_{10})^\top$.

Question 9 *(3 points)* Assuming $f \sim GP(0, k)$, what is the value of $\text{var}(f(x_1) - f(x_2))$ for two arbitrary points x_1 and x_2?

A $l^2(x_1 + x_2)^2$
C $l^2 |x_1 - x_2|$
B $l^2 x_1^2 + l^2 x_2^2$
D $l^2(x_1 - x_2)^2$

We select parameter l^* of the kernel by maximizing marginal log-likelihood of the data, i.e., :

$$l^* = \arg\max_l \log p(\boldsymbol{y}|\boldsymbol{x}, l, \sigma^2)$$

Question 10 *(4 points)* In the end, we would like to predict the value $f(x_*)$ for an arbitrary $x_* \in \mathbb{R}$. What is the mean prediction of $f(x_*)$, i.e. $\mathbb{E}(f(x_*)|x_*, \mathcal{D}, l^*, \sigma^2)$?

Hint: The following identity might be helpful. For a vector $\boldsymbol{u} \in \mathbb{R}^n$ we have:

1. $(I + \boldsymbol{u}\boldsymbol{u}^\top)^{-1} = I - \frac{\boldsymbol{u}\boldsymbol{u}^\top}{1+\|\boldsymbol{u}\|^2}$

A $\frac{\boldsymbol{x}^\top \boldsymbol{y}}{\sigma^2 + \|\boldsymbol{x}\|^2} x_*$
C $\frac{\boldsymbol{x}^\top \boldsymbol{y}}{\sigma^2 + l^{*2}\|\boldsymbol{x}\|^2} x_*$
B $\frac{l^{*2}\boldsymbol{x}^\top \boldsymbol{y}}{\sigma^2 + \|\boldsymbol{x}\|^2} x_*$
D $\frac{l^{*2}\boldsymbol{x}^\top \boldsymbol{y}}{\sigma^2 + l^{*2}\|\boldsymbol{x}\|^2} x_*$

2.1 Short questions

In this section, we discuss Gaussian Processes for regression. Assume the prior on the regression is a GP denoted by

$$f(x) \sim GP(0, k_\theta(x, x')),$$

where k_θ represents a kernel function, **not** necessarily linear. We first train the model on a dataset $\mathcal{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, $x_i \in \mathbb{R}^d, y \in \mathbb{R}$, where

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \overset{\text{i.i.d}}{\sim} \mathcal{N}(0, 1)$$

by maximizing the marginal log-likelihood $\theta_* = \arg\max_\theta \log p(\boldsymbol{y}|\boldsymbol{x}, \theta)$. Also, let $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$ and $\boldsymbol{y} = (y_1, \ldots, y_n)^\top$. We would then like to calculate $p(f(x_*)|x_*, \mathcal{D}, \theta_*)$ for $x_* \in \mathbb{R}^d$.

Question 11 *(1 point)* For any choice of the kernel function k_θ, and every dataset $\mathcal{D}$, the optimization problem $\max_\theta \log p(\boldsymbol{y}|\boldsymbol{x}, \theta)$ has only one local optimum that is also the global optimum.

A True B False

Question 12 *(1 point)* The closed form formula for $p(f(x_*)|x_*, \mathcal{D}, \theta_*)$ might not exist in general and one needs to use approximation methods such as variational inference to estimate $p(f(x_*)|x_*, \mathcal{D}, \theta_*)$.

A True B False

Question 13 *(1 point)* When the closed form formula for $p(f(x_*)|x_*, \mathcal{D}, \theta_*)$ exists, we can compute $p(f(x_*)|x_*, \mathcal{D}, \theta_*)$ with $\mathcal{O}(n^3)$ operations.

A True B False

Question 14 *(1 point)* The posterior variance $\mathrm{var}\left(f(x_*)|x_*, \mathcal{D}, \theta_*\right)$ does not depend on $\boldsymbol{y}$.

A True B False

Question 15 *(1 point)* Let $p(w)$ be the density of an arbitrary probability distribution on $\mathbb{R}^d$ ($w \in \mathbb{R}^d$). Then the kernel

$$k(x, y) = \int_{\mathbb{R}^d} p(w) e^{iw^\top (x-y)} dw$$

is isotropic.

A True B False

Question 16 *(1 point)* The computational cost of inducing point methods SoR and FITC is cubic in the number of inducing points and linear in the number of data points.

A True B False

Question 17 *(1 point)* We can't treat inducing points as hyper-parameters and optimize their positions along with other parameters by maximizing the marginal log-likelihood of the data since, in that case, the positions of the inducing points always converge to the same point.

A True B False

3 Variational Inference and MCMC

3.1 Markov Chain

Consider the graph in the Figure 2, consisting of an inner polygon (black nodes) and an outer polygon (white nodes). A car is traversing the graph's nodes via the following transition probability. At any step n, for any node in the graph, the probability of traveling to any of the three neighboring nodes is $\frac{1}{3}$. Let the random variable X_n denote the node the car is on at step n. We start traversing the graph from a **fixed** inner node X_1 as depicted on Figure 2.

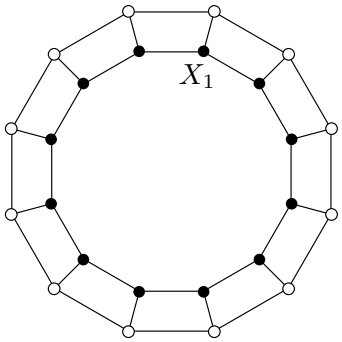


Figure 2: Related to Questions 18 - 20

Question 18 *(1 point)* The random sequence $X_1, X_2, \ldots$ is an ergodic Markov chain.

A True B False

Now consider a random variable Z_n which indicates if the car is on the outer polygon or not. Let $Z_n = 0$ if X_n is a node on the inner polygon, and $Z_n = 1$ if X_n is on the outer polygon. It can be shown that the random sequence $Z_1, Z_2, \ldots$ is a Markov chain.

Question 19 *(1 point)* The transition probability $P(Z_{n+1} = z | Z_n = z')$ does not depend on n.

A True B False

Question 20 *(3 points)* What can be said about $\lim_{n\to\infty} P(Z_n = 0)$?

A $\lim_{n\to\infty} P(Z_n = 0) = 1/2$
B $\lim_{n\to\infty} P(Z_n = 0) = 2/3$
C $\lim_{n\to\infty} P(Z_n = 0) = 3/5$
D This limit does not exist.

Next, consider a **new domain** $D = \{1, 2, 3, 4\}$ and 4 Markov chains on it. Each Markov chain $(X_1, X_2, \ldots)$ has a uniform initial distribution, i.e., X_1 is uniformly distributed on D, but every chain has a different transition matrix P. We define the transition matrix $P \in \mathbb{R}^{4\times 4}$ as $P_{i,j} = \mathbb{P}(X_{k+1} = j | X_k = i)$ for $1 \leq i, j \leq 4$.
For the next 4 questions answer the following: Does the transition matrix P guarantee ergodicity of the resulting Markov Chain?

Question 21 *(1 point)*

$$P = \begin{pmatrix} 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \end{pmatrix}$$

A True B False

Question 22 *(1 point)*

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.25 & 0.25 & 0 \\ 0.25 & 0 & 0.5 & 0.25 \\ 0 & 0 & 0.75 & 0.25 \end{pmatrix}$$

A True B False

Question 23 *(1 point)*

$$P = \begin{pmatrix} 0.5 & 0.25 & 0.25 & 0 \\ 0 & 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \end{pmatrix}$$

A True B False

Question 24 *(1 point)*

$$P = \begin{pmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 \end{pmatrix}$$

A True B False

3.2 Metropolis Hastings

MCMC algorithms can be used for approximating probability distributions of the form

$$p(x) = \frac{1}{Z} \exp\left(-f(x)\right).$$

In the lecture, we introduced the Metropolis Adjusted Langevin Algorithm (MALA) as an improved version of Metropolis Hastings with a normal Gaussian proposal distribution. MALA uses the following proposal, where $\dot{f}$ denotes the derivative with respect to x.

$$R(x'|x) = \mathcal{N}\left(x'|x - \sigma^2 \dot{f}(x), 2\sigma^2\right) \tag{1}$$

Here we will look into MALA for a specific energy function $f(x) = \frac{x^2}{2\sigma^2}$ where σ is the same constant as in Equation (1).

Question 25 *(3 points)* What is the acceptance probability α of $X_{t+1} = x'$ if $X_t = x$?

A $\min\left\{1, \exp\left\{\frac{1}{4\sigma^2}(x^2 - x'^2)\right\}\right\}$

B $\min\left\{1, \exp\left\{\left(\frac{1}{2} + \frac{1}{4\sigma^2}\right)(x^2 - x'^2)\right\}\right\}$

C $\min\left\{1, \exp\left\{\frac{3}{4\sigma^2}(x^2 - x'^2)\right\}\right\}$

D $\min\left\{1, \exp\left\{\frac{1}{4\sigma^2}(x'^2 - x^2)\right\}\right\}$

E $\min\left\{1, \exp\left\{\left(\frac{1}{2} + \frac{1}{4\sigma^2}\right)(x'^2 - x^2)\right\}\right\}$

Question 26 *(1 point)* Recall that $p(x) = \frac{1}{Z}\exp(-x^2/2\sigma^2)$, and X_t is the Markov chain induced by MALA. Then regardless of the location of the initial point X_1, the algorithm will converge to $p(x)$, i.e., for all x, the error $\lim_{t\to\infty} |P(X_t = x) - p(x)|$ will be arbitrary small.

A True B False

3.3 Variational Inference

Question 27 *(1 point)* Consider a distribution p and the variational family $Q = \{q_1, q_2\}$, illustrated in Figure 3. We have $q_1 = \arg\min_{q \in Q} \mathrm{KL}(q||p)$.

A True B False

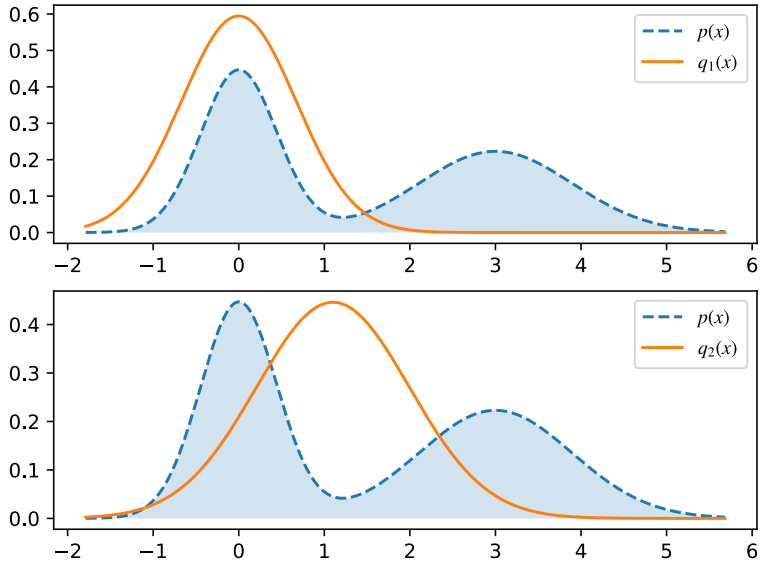


Figure 3: Related to Question 27

Question 28 *(1 point)* Let Q denote the space of two-dimensional Gaussian distributions which take the form $\mathcal{N}(0, \sigma^2 I)$, with $\sigma > 0$. Let $p(x) = \mathcal{N}(x|0, \Sigma)$ also be a two-dimensional distribution where

$$\Sigma = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}$$

for $0 < \sigma_1 < \sigma_2$. Let σ_*^2 denote the variance of the variational distribution $q_* = \arg\min_{q \in Q} \mathrm{KL}(q||p)$. Then $\sigma_* \leq \sigma_1 < \sigma_2$.

A True B False

Question 29 *(1 point)* Consider the probability density function $p(x)$ given in 4. The Laplace approximation for this distribution will also have two modes.

A True B False

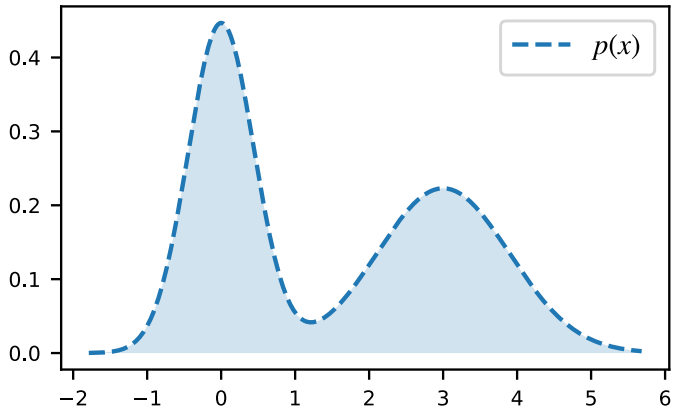


Figure 4: Related to Question 29

Question 30 *(4 point)* Let $\sigma(x) = \max(0, x)$ denote the ReLU function, and let $w > 0$. Using the reparametrization trick, calculate

$$\nabla_\mu \mathbb{E}_{x\sim\mathcal{N}(\mu,1)}\sigma(wx).$$

We denote the CDF of a normal Gaussian random variable $x \sim \mathcal{N}(0, 1)$ by $F(a) = \mathbb{P}(x \leq a)$.

A $w(1 - F(\mu))$
B $wF(\mu)$
C $wF(-\mu)$
D $w(1 - F(w\mu))$
E $wF(w\mu)$

4 Bayesian Deep Learning and Active Learning

For the following questions we consider a physical system that can be modeled by a simple Bayesian neural network. The output of the network $y_i \in \mathbb{R}$ to an input $x_i \in \mathbb{R}^3$ is described by

$$y_i = \theta^\top \sigma(\mathbf{\Lambda} x_i) + \exp(\alpha)\epsilon_i, \tag{2}$$

where $\epsilon_i \sim \mathcal{N}(0,1)$ is a random normal variable, independent of everything else, and θ and α are *unknown* parameters to be learnt. The value of $\mathbf{\Lambda} \in \mathbb{R}^{2\times 3}$ is *known*. The function σ corresponds to the sigmoid activation function that is defined as

$$\sigma(x) = \frac{1}{1+\exp(-x)}$$

and is applied to each element of x. The prior probability distributions and the respective dimensions of the *unknown* and *known* parameters are summarized below.

$\theta \in \mathbb{R}^2$	$\theta \sim \mathcal{N}(0, \mathbf{I}_{2\times 2})$
$\alpha \in \mathbb{R}$	$\alpha \sim \mathcal{N}(2,1)$, independent of θ
$\mathbf{\Lambda} \in \mathbb{R}^{2\times 3}$	deterministic and known

We can set the system to a specific configuration $\widetilde{x}$ and then observe corresponding output, which we denote $y(\widetilde{x})$. The random value $y(\widetilde{x})$ corresponds to the performance of the system

$$f_\theta(\widetilde{x}) = \theta^\top \sigma(\mathbf{\Lambda}\widetilde{x})$$

corrupted by zero-mean noise $\exp(\alpha)\epsilon$.

4.1 Bayesian Deep Learning (9 points)

Question 31 *(1 point)* Does it hold that $\mathrm{var}(y(x)|\theta) = \mathrm{var}(y(x))$ for all x?

A True B False

Question 32 *(4 points)* Given the data $\mathcal{D} = \{(x_1, y_1), \ldots, (x_{50}, y_{50})\}$, which of the following optimization problems leads to the MAP solution for the model described by Equation (2)?

A $\arg\min_{\theta,\alpha} \left(\sum_{i=1}^{50} \frac{(y_i - \theta^\top \sigma(\mathbf{\Lambda} x_i))^2}{\exp(\alpha)^2}\right) + 100\alpha + \|\theta\|_2^2 + (\alpha-2)^2$

B $\arg\min_{\theta,\alpha} \frac{1}{50}\left(\sum_{i=1}^{50} \frac{(y_i - \theta^\top \sigma(\mathbf{\Lambda} x_i))^2}{\exp(\alpha)^2}\right) + 100\alpha + \|\theta\|_2^2 + (\alpha-2)^2$

C $\arg\min_{\theta,\alpha} \frac{1}{50}\left(\sum_{i=1}^{50} (y_i - \theta^\top \sigma(\mathbf{\Lambda} x_i))^2\right) + \|\theta\|_2^2 + \alpha^2$

D $\arg\min_{\theta,\alpha} \left(\sum_{i=1}^{50} (y_i - \theta^\top \sigma(\mathbf{\Lambda} x_i))^2\right) + \|\theta\|_2^2 + \alpha^2$

E $\arg\min_{\theta,\alpha} \left(\sum_{i=1}^{50} \frac{(y_i - \theta^\top \sigma(\mathbf{\Lambda} x_i))^2}{\exp(\alpha)^2}\right) + \log(\alpha^2) + \|\theta\|_2^2 + (\alpha-2)^2$

The following applies to questions 33 and 34. We choose to approximate the posterior with a mean-field multivariate normal approximation. In other words, we assume that the approximate posterior q factorizes into $q(\theta,\alpha) = q_\theta(\theta)q_\alpha(\alpha)$, and the factors are itself normally distributed as $q_\alpha(\alpha) = \mathcal{N}(\alpha|\mu_\alpha, \Sigma_\alpha)$ and $q_\theta(\theta) = \mathcal{N}(\theta|\mu_\theta, \Sigma_\theta)$.

Question 33 *(4 points)* In order to fit our approximate posterior we use variational inference and maximize the evidence lower bound (ELBO):

$$\arg\max_{\mu_\alpha, \Sigma_\alpha \succeq 0, \mu_\theta, \Sigma_\theta \succeq 0} \mathrm{ELBO}(\mu_\alpha, \Sigma_\alpha, \mu_\theta, \Sigma_\theta).$$

Which of the expressions below corresponds to the ELBO objective for the given problem?

$$\text{ELBO}(\mu_\alpha, \Sigma_\alpha, \mu_\theta, \Sigma_\theta) = \mathbb{E}_q\left(\log(p(\mathcal{D}|\theta, \alpha))\right) + C,$$

where $p(\mathcal{D}|\theta, \alpha)$ denotes the likelihood. Please identify the correct term C.

A $C = -\frac{1}{2}\left[\left(\text{tr}(\Sigma_\theta + \mu_\theta\mu_\theta^\top) - \log\det(\Sigma_\theta)\right) + \left(\Sigma_\alpha + \mu_\alpha^2 - \log(\Sigma_\alpha)\right) - 4\mu_\alpha\right]$

B $C = -\frac{1}{2}\left[\left(\text{tr}(\Sigma_\theta + \mu_\theta\mu_\theta^\top) - \log\det(\Sigma_\theta)\right) + \left(\Sigma_\alpha + \mu_\alpha^2 - \log(\Sigma_\alpha)\right)\right]$

C $C = -\frac{1}{2}\left[\left(\text{tr}(\Sigma_\theta^{-1}(\mathbf{I}_{2\times 2} + \mu_\theta\mu_\theta^\top)) + \log\det(\Sigma_\theta)\right) + \left(\Sigma_\alpha^{-1}(1 + \mu_\alpha^2) + \log(\Sigma_\alpha)\right) - 4\mu_\alpha\Sigma_\alpha^{-1} + 4\Sigma_\alpha^{-1}\right]$

D $C = -\frac{1}{2}\left[\left(\text{tr}(\Sigma_\theta^{-1}(\mathbf{I}_{2\times 2} + \mu_\theta\mu_\theta^\top)) + \log\det(\Sigma_\theta)\right) + \left(\Sigma_\alpha^{-1}(1 + \mu_\alpha^2) + \log(\Sigma_\alpha)\right)\right]$

Hint: While solving this exercise you might find the following identity about the Kullback-Leibler divergence of two d dimensional normal distributions useful:

$$D_{KL}(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \frac{1}{2}\left(\text{tr}(\Sigma_2^{-1}(\Sigma_1 + (\mu_2 - \mu_1)(\mu_2 - \mu_1)^\top)) - d + \log\left(\frac{\det(\Sigma_2)}{\det(\Sigma_1)}\right)\right).$$

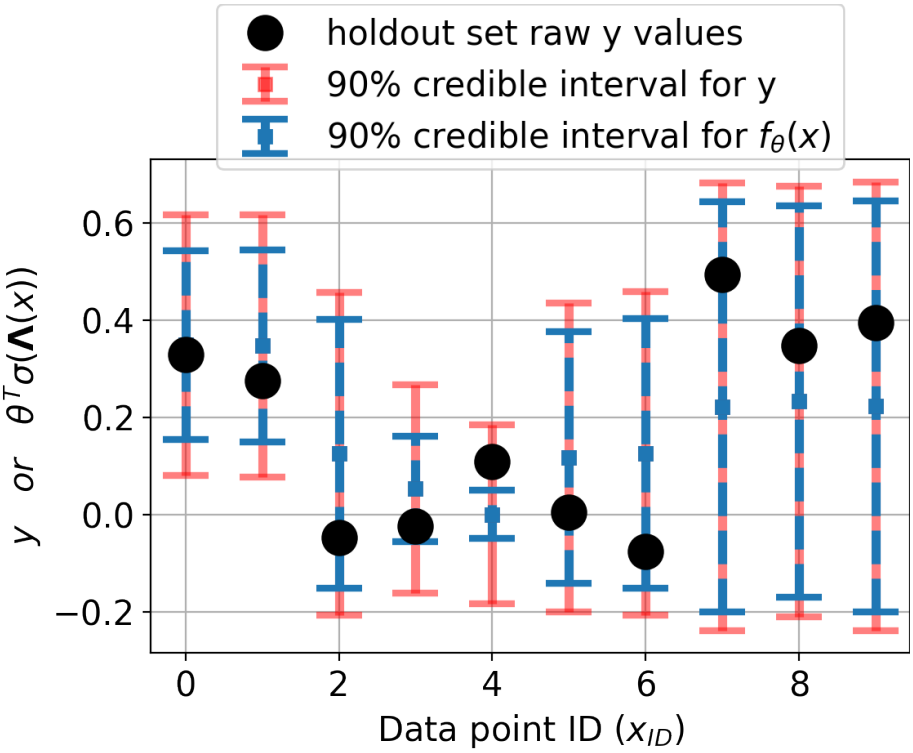
4.2 Calibration (3 points)

The questions in this subsection concern the variational approximation from the previous question.

Question 34 *(1 point)* Under the variational approximation the *posterior* predictive distribution of $f_\theta(x)$ is equal to $\mathcal{N}(\mu_\theta\sigma(\mathbf{\Lambda}x), \sigma(\mathbf{\Lambda}x)^\top\Sigma_\theta\sigma(\mathbf{\Lambda}x) + \Sigma_\alpha)$.

A True B False

Question 35 *(2 points)* We would like to evaluate the calibration of the obtained approximate posterior. In order to do so, we plot the 90% predictive *posterior* credible sets of $f_\theta(x)$ (an interval such that 90% of the probability mass is within it) for 10 different values of x_i in the holdout set in blue. We include the credible sets for posterior distribution of $y(x)$ in red, and observed samples y_i in the holdout set corresponding to x_i (black). We are interested in the empirical coverage, i.e., the fraction of samples in the holdout set that are covered by the credible sets due to the approximate posterior. Which of the following statements summarizes the conclusion we can reach from the diagram?



A The empirical coverage of y is 100%. The empirical coverage of f_θ is 90%

B The empirical coverage of y is 90%. The empirical coverage of f_θ is 100%

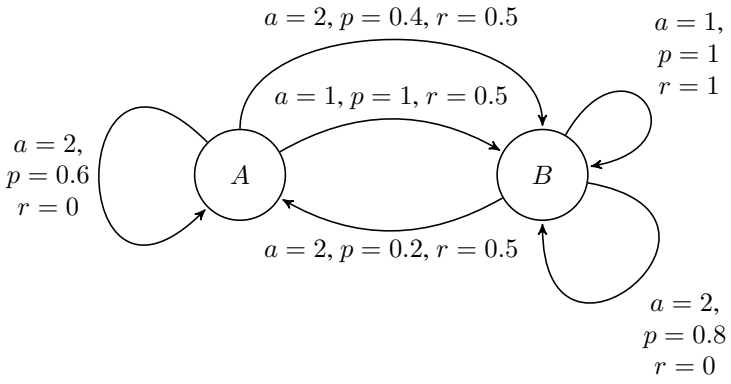
C The empirical coverage of y is 100%. The empirical coverage of f_θ cannot be inferred.

D The empirical coverage of y cannot be infered. The empirical coverage of f_θ is 90%.

5 Reinforcement Learning

5.1 Markov Decision Processes

Consider the following Markov decision process (MDP) with two states $\mathcal{X} = \{A, B\}$ and two actions $\mathcal{A} = \{1, 2\}$ with discount factor γ:



Question 36 *(1 point)* The environment has deterministic dynamics.

A True B False

Question 37 *(1 point)* Taking action 1 in state A will never result in staying in state A.

A True B False

Question 38 *(2 points)* The transition dynamics of this MDP can be described using two transition matrices P_1 and P_2 for taking action 1 and 2 respectively, where $(P_a)_{ij} = P(x_j|x_i, a)$ with $a, i, j \in \{1, 2\}$ and $x_1 = A$, $x_2 = B$. Which two transition matrices describe the environment shown above?

A $P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, P_2 = \begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix}$

B $P_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, P_2 = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$

C $P_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, P_2 = \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix}$

D $P_1 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, P_2 = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}$

Question 39 *(2 points)* Consider a deterministic policy $\pi_1 : \mathcal{X} \to \mathcal{A}$ with $\pi_1(A) = 1$, $\pi_1(B) = 1$, and assume the discount factor is $\gamma = 0.5$ for an infinite time horizon. Compute the value function $V^{\pi_1}(A)$ in state A when following policy π_1.

A $V^{\pi_1}(A) = 1.5$

B $V^{\pi_1}(A) = 1$

C $V^{\pi_1}(A) = 0.5$

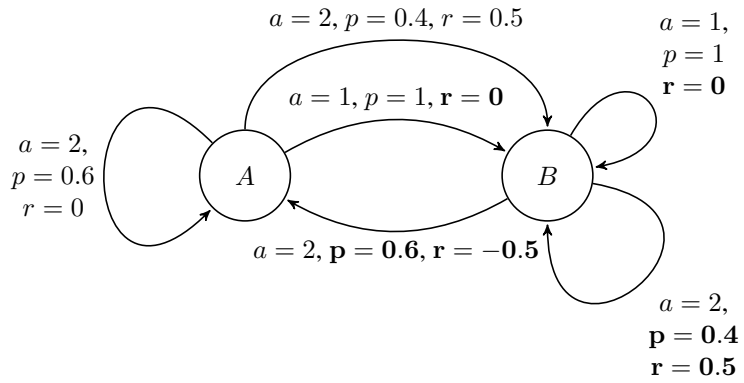
D $V^{\pi_1}(A) = 2$

Question 40 *(3 points)* Assume there is a finite time horizon T without discounting ($\gamma = 1$). Let $J(\pi) = \mathbb{E}_{P,\pi}\left(\sum_{t=0}^{T-1} r(x_t, a_t, x_{t+1})\right)$ be the expected return of policy π. Consider a deterministic policy $\pi_2 : \mathcal{X} \to \mathcal{A}$ with $\pi_2(A) = 2$, $\pi_2(B) = 2$. Assume the agent starts in either state with probability 0.5 and let $T = 2$. Compute $J(\pi_2)$.

A $J(\pi_2) = 0.32$
B $J(\pi_2) = 0.29$
C $J(\pi_2) = 0.54$
D $J(\pi_2) = 0.49$

5.2 Temporal difference learning

Consider a modified version of the MDP from the last exercise (changes are marked in bold):



Assume you are doing temporal difference (TD) learning with learning rate $\alpha = 0.2$, discount factor $\gamma = 0.5$. Your current value estimates are $\hat{V}_t(A) = 0.5$, $\hat{V}_t(B) = 1$, and you observed the following transitions:

$$t+1: \quad (x = A, a = 2, r = 0.5, x' = B)$$
$$t+2: \quad (x = B, a = 1, r = 0, x' = B)$$

Update your value estimates from these transitions one by one using bootstrapping.

Question 41 *(3 points)* After the first update your value estimates are

A $\hat{V}_{t+1}(A) = 0.6, \hat{V}_{t+1}(B) = 0.8$
B $\hat{V}_{t+1}(A) = 0.7, \hat{V}_{t+1}(B) = 1$
C $\hat{V}_{t+1}(A) = 0.6, \hat{V}_{t+1}(B) = 1$
D $\hat{V}_{t+1}(A) = 0.7, \hat{V}_{t+1}(B) = 0.8$

Question 42 *(3 points)* After the second update your value estimates are

A $\hat{V}_{t+2}(A) = 0.6, \hat{V}_{t+2}(B) = 0.9$
B $\hat{V}_{t+2}(A) = 0.7, \hat{V}_{t+2}(B) = 0.8$
C $\hat{V}_{t+2}(A) = 0.6, \hat{V}_{t+2}(B) = 0.8$
D $\hat{V}_{t+2}(A) = 0.7, \hat{V}_{t+2}(B) = 0.9$

5.3 True/False question

Question 43 *(1 point)* Policy iteration finds an optimal policy in a Markov decision process (MDP) with a number of iterations polynomial in the size of the state and action space.

A True
B False

Question 44 *(1 point)* Value iteration can sometimes find a better policy than policy iteration after convergence.

A True B False

Question 45 *(1 point)* Value iteration always converges faster (in smaller number of steps) than policy iteration.

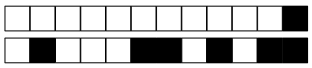
A True B False

Question 46 *(1 point)* Q-Learning is a model-free RL algorithm.

A True B False

Question 47 *(1 point)* Actor-critic algorithms use a baseline to reduce bias in the gradient estimates.

A True B False



+1/17/43+

Answer Sheet of the Probabilistic Artificial Intelligence Exam

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

⟵ Please encode your student number on the left, and write your first and last names below.

Firstname and Lastname:

.....

Question 1: A B C D
Question 2: A B C D
Question 3: A B
Question 4: A B
Question 5: A B
Question 6: A B
Question 7: A B
Question 8: A B C D E F
Question 9: A B C D
Question 10: A B C D
Question 11: A B
Question 12: A B
Question 13: A B
Question 14: A B
Question 15: A B
Question 16: A B
Question 17: A B
Question 18: A B
Question 19: A B
Question 20: A B C D
Question 21: A B
Question 22: A B
Question 23: A B
Question 24: A B
Question 25: A B C D E
Question 26: A B
Question 27: A B
Question 28: A B
Question 29: A B
Question 30: A B C D E
Question 31: A B
Question 32: A B C D E
Question 33: A B C D
Question 34: A B
Question 35: A B C D
Question 36: A B
Question 37: A B
Question 38: A B C D
Question 39: A B C D
Question 40: A B C D
Question 41: A B C D
Question 42: A B C D
Question 43: A B
Question 44: A B
Question 45: A B
Question 46: A B
Question 47: A B

For your examination, preferably print documents compiled from auto-multiple-choice.